04. Preprocessing & Feature Engineering

Understand label encoding, feature selection, scaling and synthetic sampling.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

Target encoding

LabelEncoder maps the four string labels to integer codes in alphabetical order: 0=Overweight, 1=Stunting, 2=Underweight, 3=Wasting. These numbers are identifiers, not a ranking or numeric severity scale. For multiclass algorithms, the mapping must be retained to decode predictions.

Feature selection in the notebooks

Several notebook branches drop the three income-group columns and retain Sex, Age, Height and Weight. Other branches start with all seven predictors. The final app expects exactly four columns: Sex, Age, Height and Weight. The repository does not clearly justify why the income indicators were excluded.

Scaling

StandardScaler is used in the SMOTE/SVM workflow before GridSearchCV. Standardization computes z=(x–mean)/standard deviation. It is important for SVM because distances and margin geometry are scale-sensitive. The scaler must be fit only on training data and then applied unchanged to validation/test data.

SMOTE

SMOTE synthesizes minority-class feature vectors by interpolating between a minority observation and one of its nearest minority neighbours: $x_{new} = x_i + \lambda(x_{nn} - x_i)$, where λ is between 0 and 1. It increases minority representation but can create implausible combinations if features are constrained, categorical or correlated.

Critical ordering distinction

- The file “Oversampling on the whole dataset.ipynb” applies SMOTE to all observations before splitting. This leaks information across partitions and makes downstream evaluation invalid.
- The separate SMOTE.ipynb splits first, then applies SMOTE only to X_train/y_train. This is the correct ordering in principle.
- The separate notebook uses k_neighbors=1 in one branch because the rarest class has very few examples; the main SMOTE notebook uses k_neighbors=3.

Better preprocessing pipeline

Split first. Fit all learned preprocessing (imputation, scaling, feature selection) on training data only. Apply SMOTE only inside each training fold using an imbalanced-learn Pipeline. Leave validation and test sets untouched and in their natural class proportions.